

The $1/\bar{\epsilon}$ in Row IV: what it is, and where it goes

Answering your advisor's five statements, one at a time

$$\xi = \frac{k^+}{k^+ + p^+}, \quad \bar{\xi} = 1 - \xi, \quad \xi_0 = \frac{k^+}{\sqrt{}} < 1, \quad r = y' - y, \quad L_r = \log(4\pi^2 \mu^2 r^2), \quad \frac{1}{\bar{\epsilon}} = \frac{1}{\epsilon} - \gamma_E + \log 4\pi, \quad \hat{C}(\xi) = \xi \bar{\xi} + \frac{\xi}{[1-\xi]_+} + \frac{\bar{\xi}}{\xi} = \frac{\mathcal{P}_{gg}(\xi)}{2N_c}$$

Short answer. Your advisor is right on all five points of principle, and their expression is *algebraically identical to yours* — the apparent sign difference is entirely in the reversed integration limits $\int_1^{\xi_0} = -\int_{\xi_0}^1$. But the *prescription* in their line 2 cannot be right as it stands, for a structural reason: **this row contains exactly one $1/\bar{\epsilon}$** , and that single pole multiplies $[\delta(1-\xi) \log \frac{k^+}{\Lambda} + \frac{\mathcal{P}_{gg}(\xi)}{2N_c}]$ as a whole. It therefore has *one* fate, not two. An $\overline{\text{MS}}$ fragmentation counterterm is proportional to $\mathcal{P}_{gg}(\xi)$ alone — it has no $\delta(1-\xi) \log \frac{k^+}{\Lambda}$ — so it can remove only part of the pole and must leave the rest behind, which is exactly the leftover they are objecting to. And the pole is a **large- $|z|$ (infrared, large transverse distance)** divergence, not a collinear one: the $z \rightarrow y$ region is convergent. My reading is that it cancels against the companion Group II rows through the BK identity, and that nothing should be absorbed into a fragmentation function here.

1. The advisor's expression versus yours: they are the same

Write the row's non- δ integrand with the common prefactor $\mathcal{P}_N = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{N_c}{k^+}$ stripped, and abbreviate

$$\mathcal{X} = \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2}, \quad \mathcal{R} = \frac{(x-y) \cdot r (x'-y') \cdot r}{(x-y)^2 (x'-y')^2 r^2}, \quad \mathcal{M} = \frac{1}{\bar{\epsilon}} + 1 - L_r.$$

Yours, with the measure written in the natural order $\int_{\xi_0}^1 \frac{d\xi}{\xi^2}$:

$$\mathcal{D} = \hat{C} \mathcal{M} \mathcal{X} - d_{\perp} \xi \bar{\xi} \mathcal{R} - \left[\frac{\xi}{[1-\xi]_+} + \frac{\bar{\xi}}{\xi} \right] \mathcal{X}.$$

Theirs, reading each of their lines 2, 3, 4 exactly as written, with $\frac{\mathcal{P}_{gg}}{2} = N_c \hat{C}$ and their $\frac{1}{k^+}$ in line 4 supplying the missing N_c :

$$\mathcal{A} = \underbrace{-\hat{C} \frac{1}{\bar{\epsilon}} \mathcal{X}}_{\text{their line 2}} + \underbrace{d_{\perp} \xi \bar{\xi} \mathcal{R} + \left[\frac{\xi}{[1-\xi]_+} + \frac{\bar{\xi}}{\xi} \right] \mathcal{X}}_{\text{their line 3}} - \underbrace{\hat{C} (1 - L_r) \mathcal{X}}_{\text{their line 4}}.$$

Sympy gives $\mathcal{A} + \mathcal{D} = 0$ identically. So $\mathcal{A} = -\mathcal{D}$. But every one of their ξ -integrals is written $\int_1^{k^+/\sqrt{}}$, and since $\xi_0 = k^+/\sqrt{}$ < 1 this is a *reversed* integration, $\int_1^{\xi_0} = -\int_{\xi_0}^1$. The two minus signs cancel:

$$\int_1^{\xi_0} \frac{d\xi}{\xi^2} \mathcal{A} = \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \mathcal{D} \quad \text{exactly.}$$

This is worth being explicit about with your advisor, because it is a *trap*: if anyone later reads $\int_1^{k^+/\sqrt{}}$ as if it were positively ordered, the whole non- δ sector of the cross section flips sign relative to the evolution term. I would write all four lines with $\int_{\xi_0}^1$ and put the signs on the terms.

Two further points of agreement, which are reassuring:

- Their line 1 (evolution) carries $[\frac{1}{\bar{\epsilon}} - L_r]$ with **no** +1, while their line 4 carries $[1 - L_r]$ with the +1. That asymmetry is correct and not a typo: the $\frac{1}{p^+}$ block of the row comes as $\frac{1}{p^+} \mathcal{M} \mathcal{X} - \frac{1}{p^+} \mathcal{X} = \frac{1}{p^+} (\frac{1}{\bar{\epsilon}} - L_r) \mathcal{X}$, so the +1 has already cancelled on whatever the $\frac{1}{p^+}$ pole produces — and the $\delta(1-\xi)$ of the plus prescription is produced by exactly that pole.
- Their $\log \frac{\sqrt{}}{\Lambda} + \log \frac{k^+}{\sqrt{}} = \log \frac{k^+}{\Lambda}$ is the rapidity factorisation split: the first piece is absorbed into the JIMWLK evolution of the target from Λ up to $\sqrt{}$, the second is the finite NLO remainder. That is the right thing to do and it agrees with your $\log \frac{k^+}{\Lambda}$.

One normalisation to check. Their line 2 is written as $-\frac{g^2}{8\pi^2}\frac{1}{\epsilon}\int\frac{d\xi}{\xi^2}\frac{\mathcal{P}_{gg}}{2}\frac{d^3N_{LO}}{d^3k}\left(\frac{k}{\xi}\right)$. For that to match the prefactor $\mathcal{P}_N$ carried by lines 1, 3, 4, you need $\frac{d^3N_{LO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^2}{\pi^2}\frac{1}{k^+}\int_{x,x',y,y'}\mathcal{X}\mathcal{U}$. Compare this against your LO row before you accept their line 2; if the LO normalisation is different, line 2 is inconsistent with lines 1, 3, 4 by that ratio.

2. It simplifies further, and the simplification is the whole point

Neither version is fully reduced. The $+1$ inside $\mathcal{M}$ cancels against the explicit $-\left[\frac{\xi}{[1-\xi]_+} + \frac{\bar{\xi}}{\xi}\right]\mathcal{X}$, because those are two of the three terms of $\hat{\mathcal{C}}$. Term by term, with $\hat{\mathcal{C}} = \xi\bar{\xi} + \frac{\xi}{[1-\xi]_+} + \frac{\bar{\xi}}{\xi}$,

$$\underbrace{\left[\frac{\xi}{[1-\xi]_+} + \frac{\bar{\xi}}{\xi}\right](\mathcal{M}-1)\mathcal{X}}_{+1 \text{ gone}} + \underbrace{\xi\bar{\xi}\mathcal{M}\mathcal{X} - d_\perp\xi\bar{\xi}\mathcal{R}}_{+1 \text{ survives here}} = \hat{\mathcal{C}}\left[\frac{1}{\epsilon} - L_r\right]\mathcal{X} + \xi\bar{\xi}[\mathcal{X} - d_\perp\mathcal{R}].$$

(Sympy: difference = 0.) Adding the δ sector, which carries $\frac{1}{\epsilon} - L_r$ as noted above, the entire row collapses to

$$\left.\frac{d^3N^{\text{NLO}}}{d^3k}\right|_{\text{Row IV}} = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{N_c}{k^+}\int_{\xi_0}\frac{d\xi}{\xi^2}e^{-i\frac{k}{\xi}r}\int_{x,x',y,y'}\left\{\left[\delta(1-\xi)\log\frac{k^+}{\Lambda} + \frac{\mathcal{P}_{gg}(\xi)}{2N_c}\right]\left[\frac{1}{\epsilon} - \log(4\pi^2\mu_F^2r^2)\right]\mathcal{X} + \xi(1-\xi)[\mathcal{X} - d_\perp\mathcal{R}]\right\}\mathcal{U}$$

Four lines have become two, and now the structure your advisor is worried about is visible in one glance: **there is a single $\frac{1}{\epsilon}$, and it does not belong to the $\mathcal{P}_{gg}$ term alone.** Its coefficient is $\delta(1-\xi)\log\frac{k^+}{\Lambda} + \frac{\mathcal{P}_{gg}(\xi)}{2N_c}$. The last block, $\xi(1-\xi)[\mathcal{X} - d_\perp\mathcal{R}]$, is finite, μ -independent and Λ -independent — it is the genuine NLO hard remainder.

3. The five statements, one at a time

Statement	Verdict	Why
“The second term has $1/\epsilon$ in front of it?!”	Right	It does. A $1/\epsilon$ surviving in a cross section is always either wrong or not yet finished. Here it is the latter: Row IV is one contraction of Group II, not the finished NLO answer.
“ $1/\epsilon$ needs to cancel or be absorbed somewhere.”	Right	Agreed. The sharpening is: there is only <i>one</i> pole in this row, so it gets <i>one</i> treatment. “Absorb this half, cancel that half” is not available — both halves are the same $\int d^2z$.
“This is not DGLAP.”	Right	$\frac{\mathcal{P}_{gg}(\xi)}{2\epsilon}$ is the DGLAP <i>counterterm</i> , i.e. the object you subtract. Calling it the DGLAP term is exactly backwards.
“DGLAP is finite, not $1/\epsilon$.”	Right	The DGLAP term in a factorised cross section is $\frac{\alpha_s}{2\pi}\int\frac{d\xi}{\xi^2}\mathcal{P}_{gg}(\xi)\log\frac{Q^2}{\mu_F^2}\sigma_{LO}(k/\xi)$ — finite by construction.
“DGLAP should come with some $\log\mu$.”	Right, and achievable	If you do subtract an $\overline{\text{MS}}$ counterterm $\frac{\mathcal{P}_{gg}}{2\epsilon}$, what is left is $\frac{\mathcal{P}_{gg}}{2}\left[1 - \log(4\pi^2\mu_F^2r^2)\right]$ — finite, carrying $\log\mu_F$. That is literally their line 4. So their expectation is realisable; the question is whether it is <i>legitimate</i> , which is §4.

4. Is this pole collinear? No — it is a large-distance pole

This decides everything, so do it by hand. The pole comes from

$$\mathcal{G}^{mm'} = \int d^2z \frac{(y-z)^m (y'-z)^{m'}}{(y-z)^2 (y'-z)^2},$$

which has two potentially dangerous regions.

(a) Short distance, $z \rightarrow y$ (the emitted gluon collinear to the measured one)

Put $u = y - z$, $|u| = \rho \rightarrow 0$, and use $y' - z = u + r \rightarrow r$:

$$\frac{u^m (u+r)^{m'}}{u^2 (u+r)^2} \longrightarrow \frac{u^m}{\rho^2} \cdot \frac{r^{m'}}{r^2} \sim \frac{1}{\rho} \cdot \frac{1}{|r|}, \quad d^2 z = \rho d\rho d\phi,$$

$$\implies \int_0 \rho d\rho \frac{1}{\rho} \frac{1}{|r|} = \int_0 \frac{d\rho}{|r|} \text{ finite.}$$

The single power of u^m in the numerator softens the $1/u^2$ to $1/\rho$, and the two-dimensional measure $\rho d\rho$ then wins. **There is no collinear divergence in this integral.** The same holds at $z \rightarrow y'$ by symmetry.

(b) Large distance, $|z| \rightarrow \infty$

$$\frac{(y-z)^m (y'-z)^{m'}}{(y-z)^2 (y'-z)^2} \longrightarrow \frac{z^m z^{m'}}{z^4} \sim \frac{1}{z^2}, \quad d^2 z = |z| d|z| d\phi \implies \int^R \frac{d|z|}{|z|} = \log R \text{ divergent.}$$

With a hard cutoff $|z| < R$ this is the result you already have,

$$\mathcal{G}^{mm'} \Big|_{\text{cutoff}} = \frac{\pi}{2} \left(\log \frac{R^2}{r^2} + 1 \right) \delta^{mm'} - \pi \hat{r}^m \hat{r}^{m'},$$

in which R is the *infrared* cutoff and r only sets the scale. Dimensional regularisation trades $\log R^2$ for $\frac{1}{\epsilon}$, so

The $\frac{1}{\epsilon}$ of Row IV is infrared — it comes from the emitted gluon sitting arbitrarily far away in transverse position (arbitrarily small transverse momentum). It is *not* a collinear pole. A fragmentation function absorbs collinear poles. It has no business absorbing this one.

5. What it really cancels against: the BK identity

In the CGC the large- $|z|$ region is not divergent in the full answer, because the emitted gluon carries a Wilson line $U(z)$ and $U(z) \rightarrow 1$ as $|z| \rightarrow \infty$; the real emission is then cancelled by the virtual ones. Algebraically that cancellation is the identity

$$\frac{(y-z) \cdot (y'-z)}{(y-z)^2 (y'-z)^2} - \frac{1}{2(y-z)^2} - \frac{1}{2(y'-z)^2} = -\frac{(y-y')^2}{2(y-z)^2 (y'-z)^2}.$$

(SymPy: difference = 0.) The right-hand side falls as $1/|z|^4$, so $\int d^2 z$ of it converges: the log at large $|z|$ is gone. The first term is Row IV; the two subtractions are the virtual/self-energy partners in Group II. Numerically, with $|r| = 1$ and a cutoff at $|z| = R$:

R	real term alone	real + virtual
20	18.824	−76.230
60	25.723	−75.908
180	32.641	−76.840
540	39.547	−77.705

The real term alone climbs by $\pi \log 9 = 6.9$ per column, exactly $\pi \log \frac{R^2}{r^2}$; the combination saturates. That is the pole disappearing.

There is a second, independent argument that points the same way, and it is the one I would put to your advisor. The coefficient of $\delta(1-\xi) \log \frac{k^+}{\Lambda}$ in the boxed result of §2 is, by construction, $(\text{LO}) \times (\text{the JIMWLK kernel})$: that is what a rapidity-evolution term is. The JIMWLK kernel is finite. Therefore the bracket $[\frac{1}{\epsilon} - \log(4\pi^2 \mu^2 r^2)]$ **must** become finite once the companion colour structures are restored. And because the very same bracket multiplies $\frac{\mathcal{P}_{qq}(\xi)}{2N_c}$, it becomes finite there at the same moment, for free.

6. Why the half-and-half prescription is not self-consistent

Suppose one insisted on their line 2 — absorb $\frac{\mathcal{P}_{qq}}{2\epsilon}$ into a gluon fragmentation function. Then:

1. The $\delta(1 - \xi) \log \frac{k^+}{\Lambda} \cdot \frac{1}{\epsilon}$ piece is untouched, because no $\overline{\text{MS}}$ fragmentation counterterm contains a rapidity logarithm $\log \frac{k^+}{\Lambda}$. It would still have to cancel against the companion rows.
2. But it is *the same* $\int d^2 z$. If that z -integral's divergence cancels against the companion rows for the $\delta(1 - \xi)$ piece, it cancels for *every* ξ — the z -integral does not know about ξ at all.
3. So the counterterm in line 2 would be subtracting a pole that is already cancelling: a double count, leaving a spurious finite $-\frac{\mathcal{P}_{gg}}{2} \times (\text{scheme})$ behind.

The logic closes: the ξ -independence of the z -integral is what forbids the split.

7. What I would say back

- **Agree with the principle.** A $\frac{1}{\epsilon}$ in a final answer is not acceptable, and $\frac{\mathcal{P}_{gg}}{\epsilon}$ is a counterterm, not DGLAP. Both of those are right.
- **Show the compact form of §2.** It makes the single-pole structure manifest and it is four lines shorter than either version. Their line 3 and the $+1$ in their line 4 cancel each other.
- **State where the pole lives.** $\int d^2 z$ at large $|z|$, and give the two-line power counting of §4(a)–(b) showing the $z \rightarrow y$ region is convergent. This is the fact that settles “absorb or cancel”, and it is checkable in thirty seconds.
- **Propose the check instead of asserting the answer.** The honest position is that Row IV alone cannot decide its own fate: the companion Group II contractions (the ones that keep $U(z)$ inside the z -integral, and the virtual/self-energy rows) have not been computed yet. Compute two of them and test the BK identity of §5 on the sum. If the $\log R^2$ cancels, the pole was never collinear and no fragmentation function is needed at this order in this row.
- **Where $\log \mu_F$ will come from.** After the cancellation, $\mathcal{P}_{gg}(\xi)$ multiplies $\frac{1}{\pi} \int d^2 z \frac{-(y-y')^2}{2(y-z)^2(y'-z)^2}$ — finite, μ -independent, a logarithm of a ratio of dipole sizes rather than of μ_F . That is the CGC form of the collinear logarithm: $\log \frac{1}{r^2 \mu_F^2}$ only appears once you additionally factorise the small- r region into a fragmentation function at scale μ_F , which is a separate step, done after Group II is summed and not before.

What I cannot prove from here. That the cancellation actually goes through. Row IV is one colour contraction; the statement “the pole cancels” is a statement about the sum over Group II, and the other members are not yet in hand. Everything above — the identity of the two expressions, the collapse to one pole, the infrared (not collinear) nature of that pole, the BK identity and its numerics — is checked. The fate is an argued expectation, not a checked result. Say so; it is a stronger position than either “it cancels” or “absorb it” asserted flat.

Checks: `src/rowG2_advisor.py` (§1 term-by-term identity, the counterterm normalisation, the single-pole coefficient, the power counting of §4), `src/rowG2_fate.py` (§2 collapse, the BK identity and its cutoff numerics, the $\overline{\text{MS}}$ fragmentation subtraction), `src/rowG2_v5.py` (the chain (0.25)→(0.26)→(0.27) this rests on).